

6-5 Tracking Error Circuit and Kick, Hold Circuits

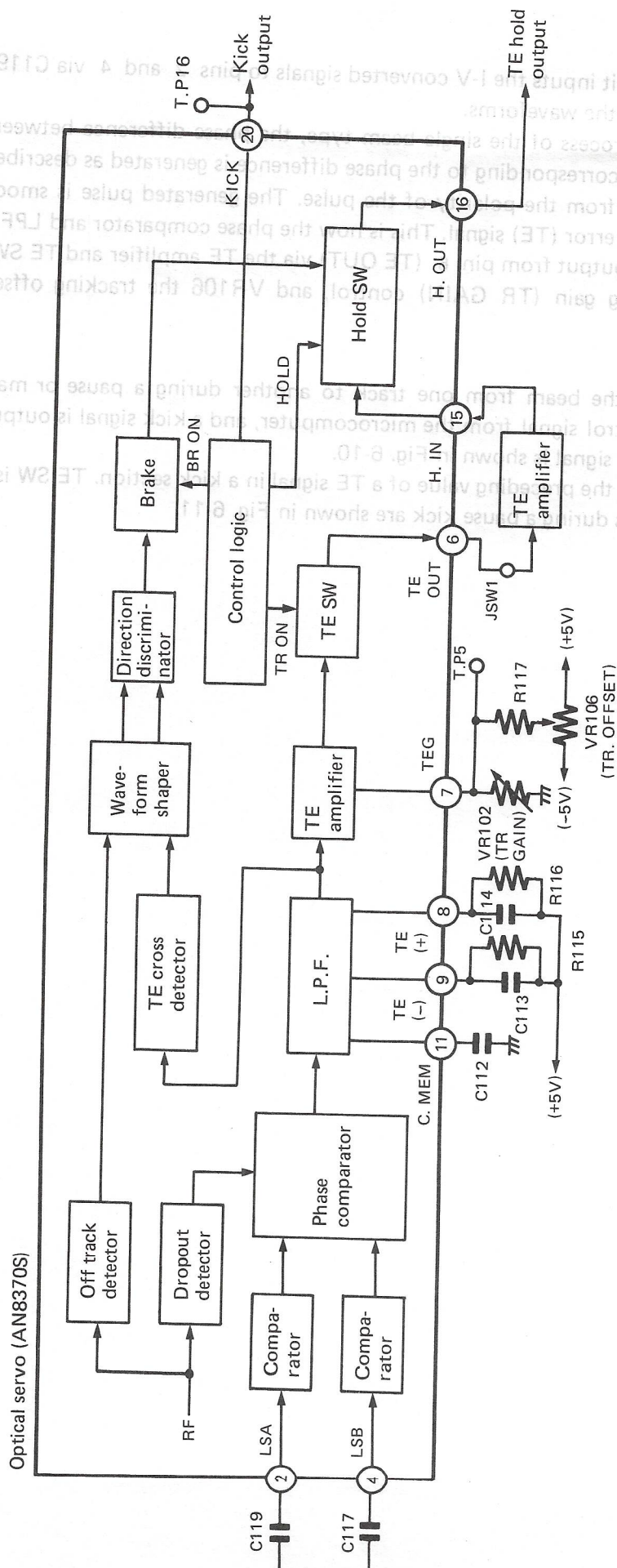


Fig. 6-9 Tracking Error Circuit and Kick, Hold Circuits

(1) Tracking error circuit

The tracking error circuit inputs the I-V converted signals to pins 2 and 4 via C119 and C117.

The comparators shape the waveforms.

In the tracking error process of the single beam type, the phase difference between the two signals is detected and a pulse corresponding to the phase difference is generated as described in 2-3. Deviating direction is judged from the polarity of the pulse. The generated pulse is smoothed by the LPF into an analog tracking error (TE) signal. This is how the phase comparator and LPF operate.

Then, the TE signal is output from pin 6 (TE OUT) via the TE amplifier and TE SW.

VR102 is the tracking gain (TR GAIN) control, and VR106 the tracking offset (TR OFFSET) control.

(2) Kick and hold circuits

Kick means jumping the beam from one track to another during a pause or manual search. Kick timing is set by a control signal from the microcomputer, and a kick signal is output to pin 20. The waveform of each kick signal is shown in Fig. 6-10.

The hold circuit holds the preceding value of a TE signal in a kick section. TE SW is kept off during a kick. Signal waveforms during a pause kick are shown in Fig. 6-11.

Fig. 6-8 Tracking Error Circuit and Kick Hold Circuit



Fig. 6-9 Kick and Hold Circuit

6-4 Auto Focus Circuit and Focus Error Circuit

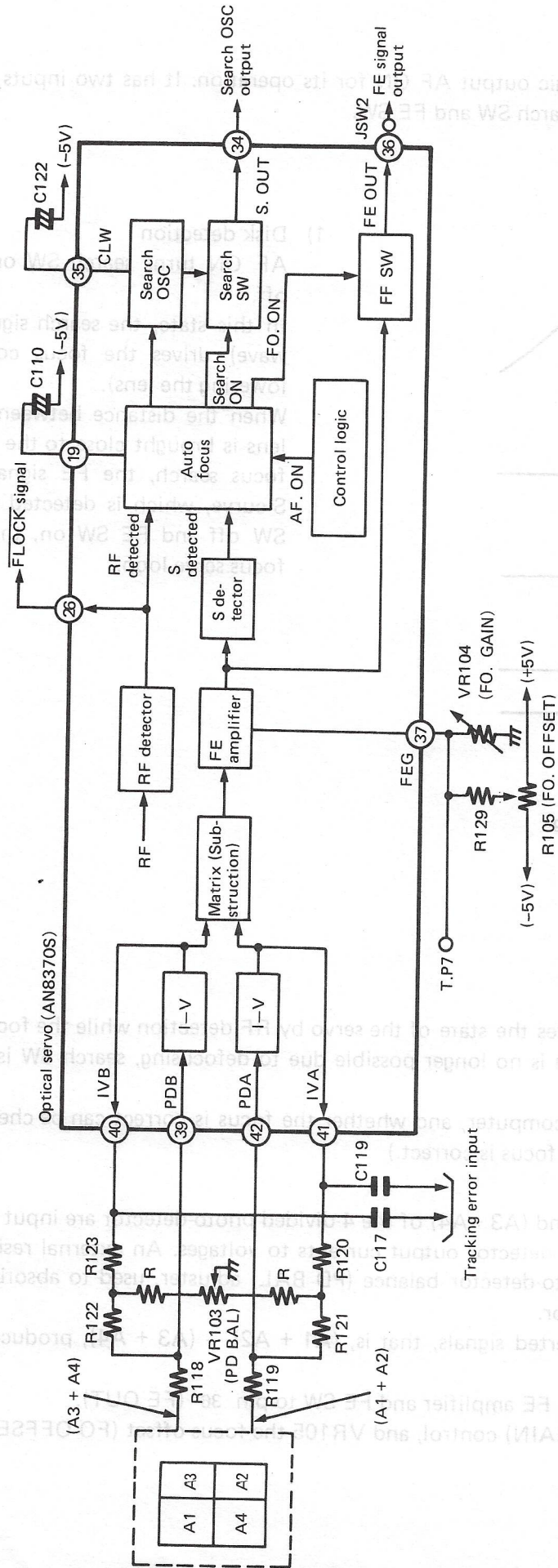


Fig. 6-7 Focus Error Circuit and Auto Focus Circuit

(1) Auto focus circuit

The auto focus uses control logic output AF ON for its operation. It has two inputs, RF detection and S detection, and controls search SW and FE SW.

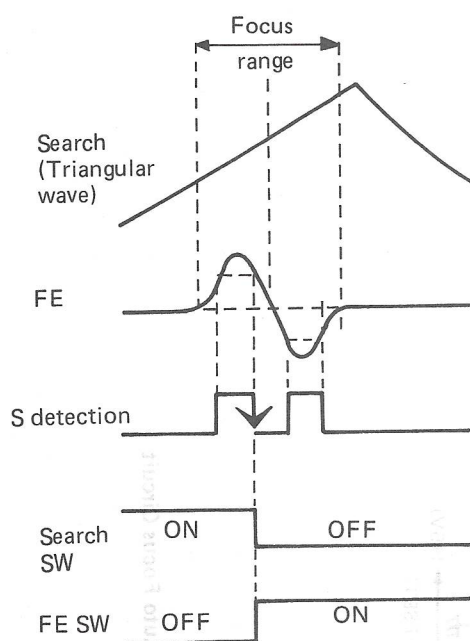


Fig. 6-8 Fo ON Timing

1) Disk detection

AF ON turns search SW on and FE SW off.

In this state, the search signal (triangular wave) drives the focus coil (raising or lowering the lens).

When the distance between the disk and lens is brought close to the focal point by focus search, the FE signal develops an S-curve, which is detected to turn search SW off and FE SW on, thus closing the focus serve loop.

2) Re-focusing

The auto focus discriminates the state of the servo by RF detection while the focus servo loop is closed. When RF detection is no longer possible due to defocusing, search SW is turned to start focus search.

This is input to the microcomputer, and whether the focus is correct can be checked at pin 26 (F LOCK). (Low when the focus is correct.)

(2) Focus error circuit

The diagonal sums $(A1 + A2)$ and $(A3 + A4)$ of the 4-divided photo-detector are input to pins 39 and 42. I-V converts the photo-detector output currents to voltages. An external resistor is used for gain setting. VR101 is a photo-detector balance (PD BAL) adjuster, used to absorb the sensitivity difference of the photo-detector.

Subtraction of the two converted signals, that is, $(A1 + A2) - (A3 + A4)$, produces a focus error signal (FE).

This FE signal is routed via the FE amplifier and FE SW to pin 36 (FE OUT).

VR104 is the focus gain (FO GAIN) control, and VR105 the focus offset (FO OFFSET) control.

6-2 Laser APC Circuit

Fig. 6-3 shows the laser APC circuit.

When a laser ON mode command is emitted, pin 5 goes high to turn the laser on.

A loop circuit is formed by Q and D (the laser paired with a PIN diode) and the APC in the IC, and laser beam intensity is detected by the PIN diode in APC operation.

If laser beam intensity increases, the PIN diode's photo-electric current increases so that the pin 10 voltage rises (pin 12 voltage is the reference voltage) to decrease laser beam intensity.

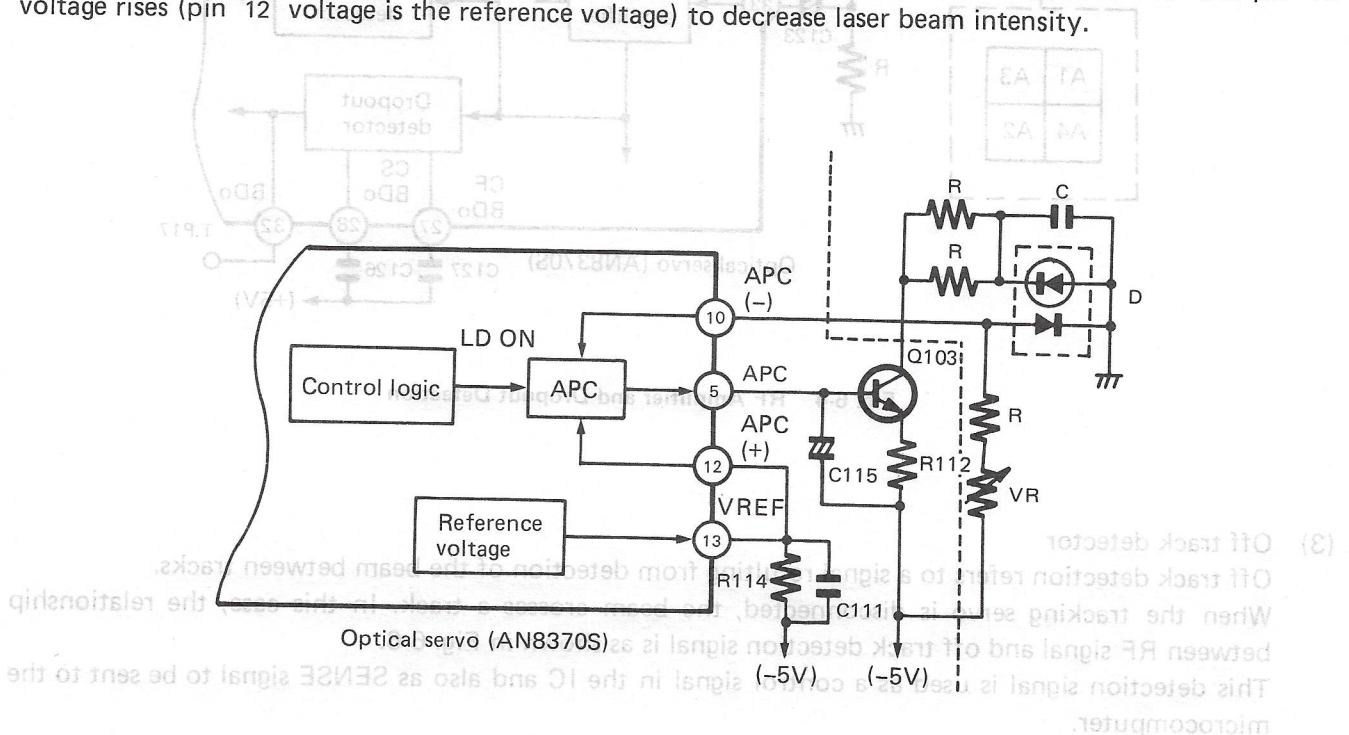


Fig. 6-3 Laser APC Circuit

6-3 RF Amplifier and Dropout Detector Circuits

(1) RF amplifier circuit

RF signal is the total sum of the 4-divided photo-detector ($A1 + A2 + A3 + A4$) of the optical pickup. These are added in the optical pickup, and are input via Q to pin 33. (Q is located in the optical pickup.)

The output is taken out from pin 31, from which it is sent to the next IC for PLL (AN8371S).

The gain between pin 33 and pin 31 is about 34 dB.

(2) Dropout detector circuit

In dropout detection, a dropout on the disk is detected from the above-mentioned RF signal, the tracking servo is stabilized, and its detection signal is output to pin 32.

The detecting process is shown in Fig. 6-5. The level goes high if a dropout is detected.

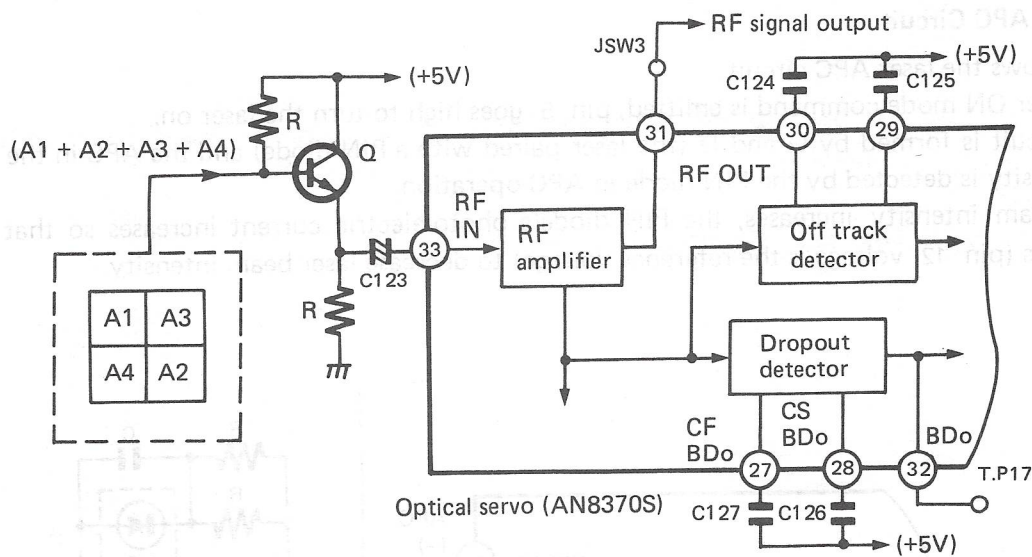


Fig. 6-4 RF Amplifier and Dropout Detection

(3) Off track detector

Off track detection refers to a signal resulting from detection of the beam between tracks.

When the tracking servo is disconnected, the beam crosses a track. In this case, the relationship between RF signal and off track detection signal is as shown in Fig. 6-6.

This detection signal is used as a control signal in the IC and also as SENSE signal to be sent to the microcomputer.

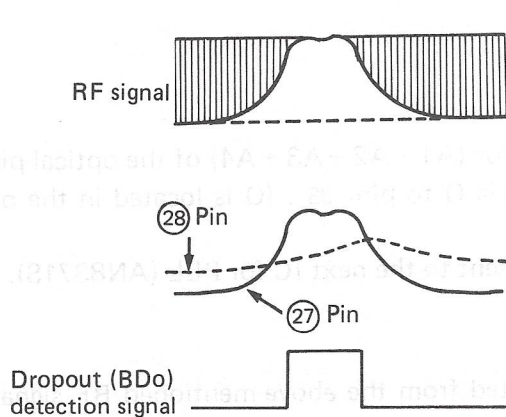


Fig. 6-5 Dropout Detection

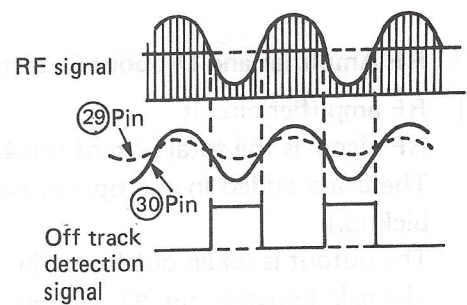


Fig. 6-6 Off Track Detection

6-1 Control Logic Circuit

Control signals CNT1 to CNT4 are 4-bit parallel output signals from the microcomputer. The logic circuit of AN8370S decodes these control signals and outputs the required control signals to the optical servo system.

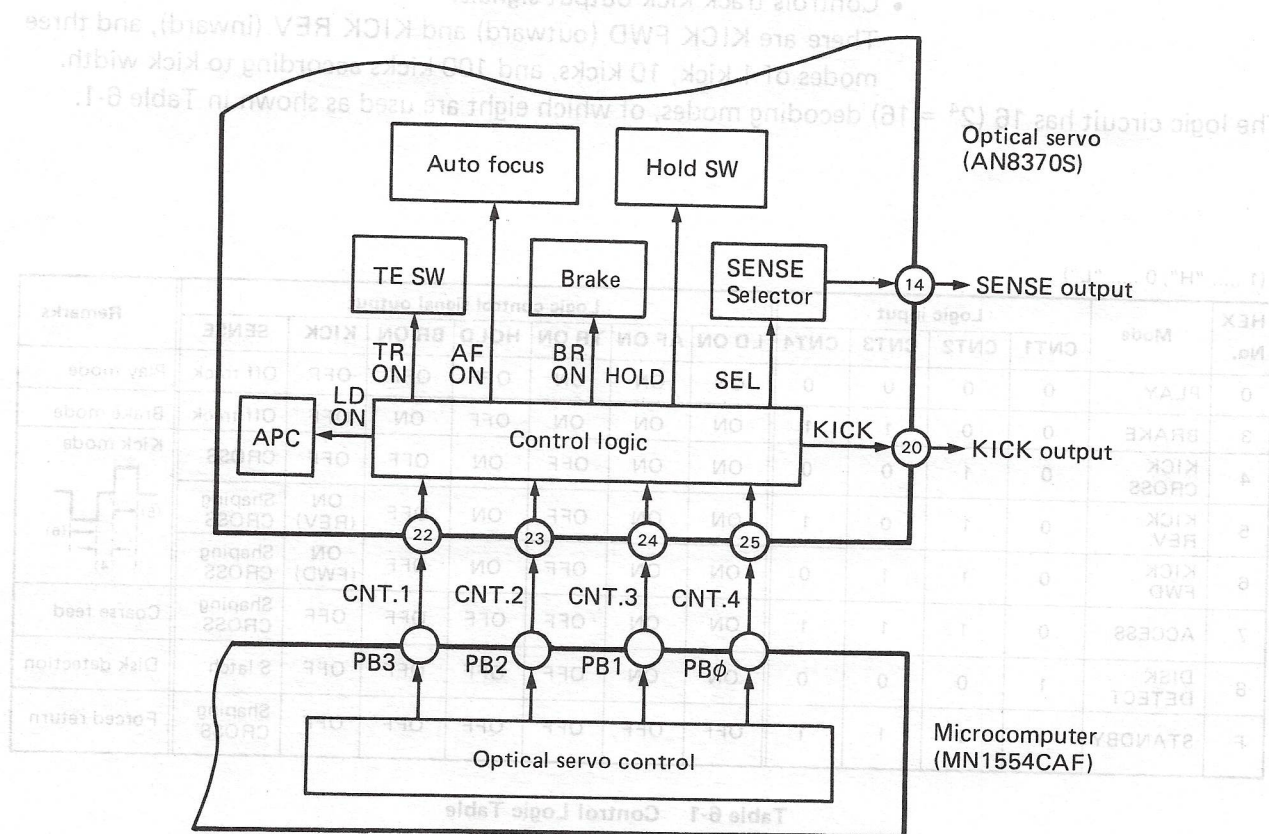


Fig. 6-2 Control Logic Circuit

- (1) Control signals and their functions (See Fig. 6-2.)
- 1) LD ON Laser ON command
 - Controls laser APC ON/OFF.
 - 2) AF ON Auto focus command
 - Controls the focus servo loop to open or close, and the search OSC switch ON/OFF.
 - 3) TR ON Tracking servo command
 - Controls the tracking servo loop switch.
 - 4) HOLD Tracking hold command
 - Holds tracking error at preceding value during track kick.
 - 5) BR ON Tracking brake command
 - After high-speed feed or kick, brakes the tracking actuator to absorb unnecessary vibration in a short time to facilitate tracking.

6) SEL SENSE signal select command

- Selects SENSE signal (that indicates optical servo status) to be sent from AN8370S to the microcomputer.
SENSE signal changes depending on the system mode.

7) KICK Track kick command

- Controls track kick output signals.

There are KICK FWD (outward) and KICK REV (inward), and three modes of 1 kick, 10 kicks, and 100 kicks according to kick width.

The logic circuit has $16 (2^4 = 16)$ decoding modes, of which eight are used as shown in Table 6-1.

(1 "H", 0 "L")

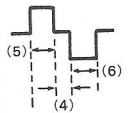
HEX No.	Mode	Logic input				Logic control signal output							Remarks
		CNT1	CNT2	CNT3	CNT4	LD ON	AF ON	TR ON	HOLD	BR ON	KICK	SENSE	
0	PLAY	0	0	0	0	ON	ON	ON	OFF	OFF	OFF	Off track	Play mode
3	BRAKE	0	0	1	1	ON	ON	ON	OFF	ON	OFF	Off track	Brake mode
4	KICK CROSS	0	1	0	0	ON	ON	OFF	ON	OFF	OFF	CROSS	Kick mode 
5	KICK REV	0	1	0	1	ON	ON	OFF	ON	OFF	ON (REV)	Shaping CROSS	
6	KICK FWD	0	1	1	0	ON	ON	OFF	ON	OFF	ON (FWD)	Shaping CROSS	
7	ACCESS	0	1	1	1	ON	ON	OFF	OFF	OFF	OFF	Shaping CROSS	Coarse feed
8	DISK DETECT	1	0	0	0	ON	ON	OFF	OFF	OFF	OFF	S latch	Disk detection
F	STANDBY	1	1	1	1	OFF	OFF	OFF	OFF	OFF	OFF	Shaping CROSS	Forced return

Table 6-1 Control Logic Table

(1) LCD display

The LCD is used for track and time display.

Serial data synchronized with the clock is input from the microcomputer to the LCD driver, and each time that serial data ends, LOAD signal is added to the data to be sent to the LCD.

OSC is a terminal for generating a COM frequency, which is one-eighth of the OSC frequency. Signal voltage between COM and each SEG are used for LCD display as follows:

SEG output with phase opposite to COM signal: ON

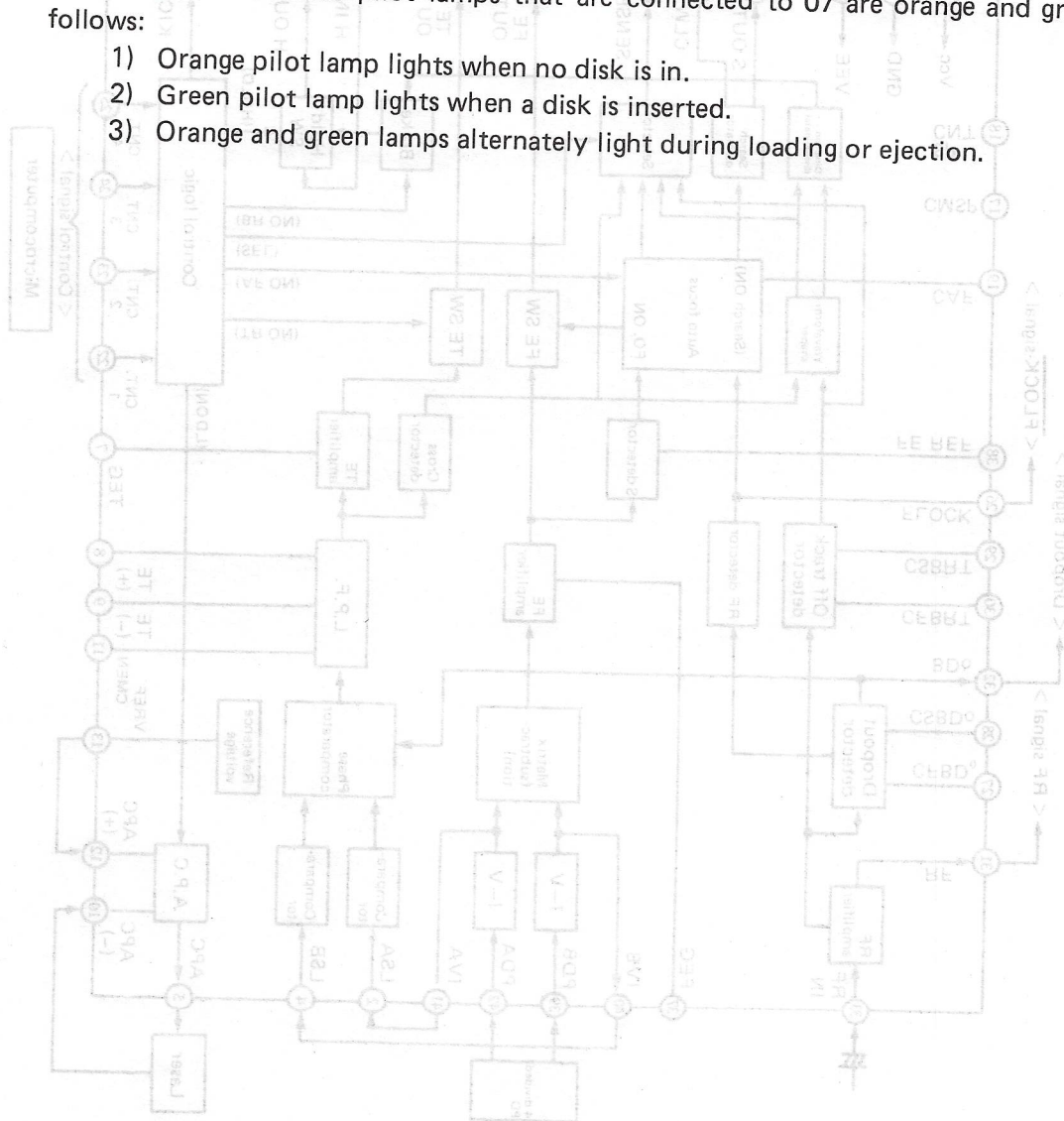
```
SEG output with same phase to COM signal ..... OFF
```

(2) LED display

The LED display indicates seven statuses.

Serial data synchronized with the clock is input from the microcomputer to the LED driver, and each time that serial data ends, LATCH signal is added to the data for selecting an LED and giving an indication. The two pilot lamps that are connected to 07 are orange and green and operate as follows:

- 1) Orange pilot lamp lights when no disk is in.
- 2) Green pilot lamp lights when a disk is inserted.
- 3) Orange and green lamps alternately light during loading or ejection.



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6. AN8370S CIRCUIT OPERATION

The optical servo reads the signals recorded on a disk using the laser beam. AN8370S is the main IC for the optical servo, consisting of laser APC, (including auto focus), tracking servo (including kick), drop output detector and controls the optical servo in this IC.

A block diagram of AN8370S is shown in Fig. 6-1.

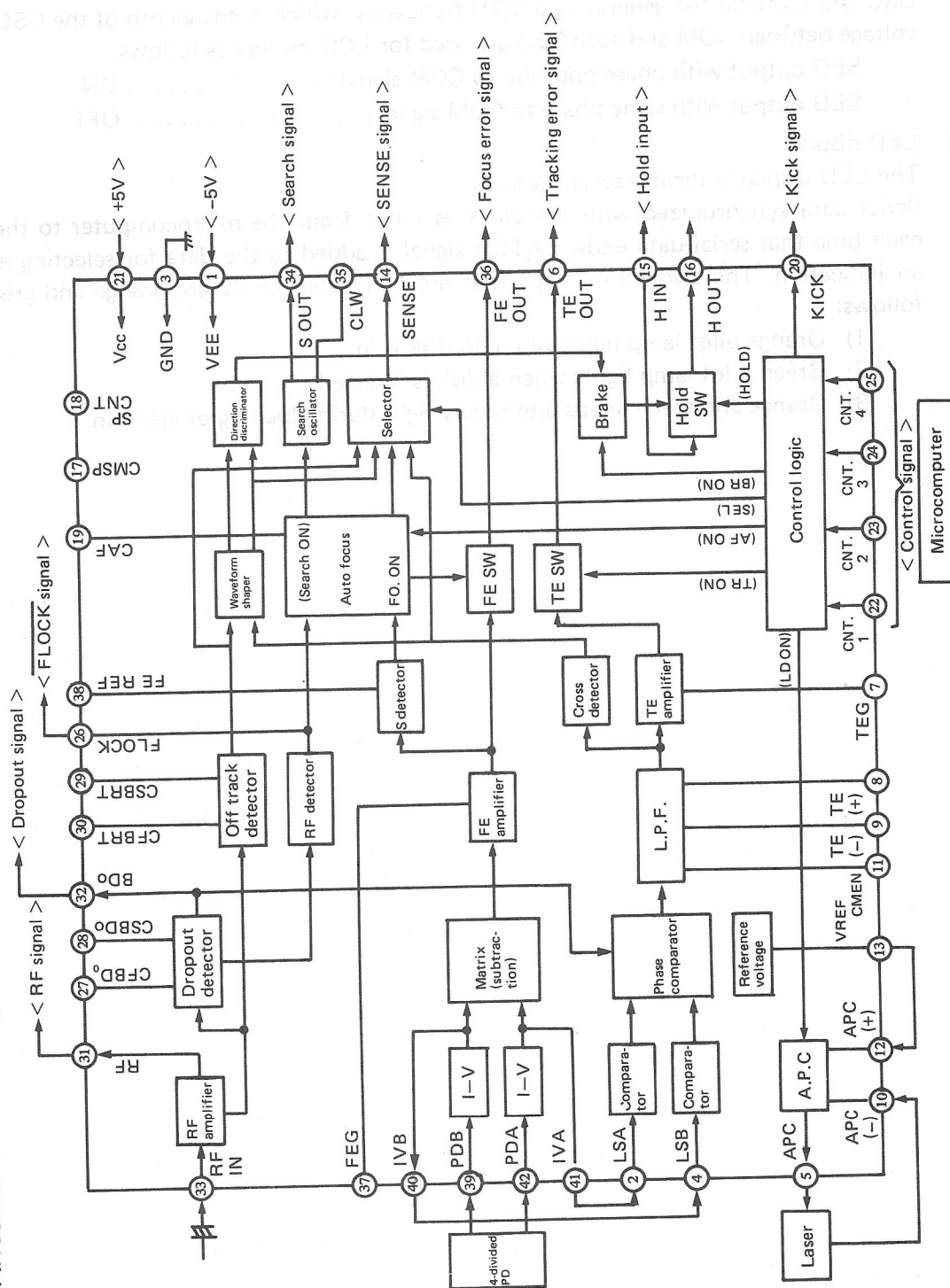
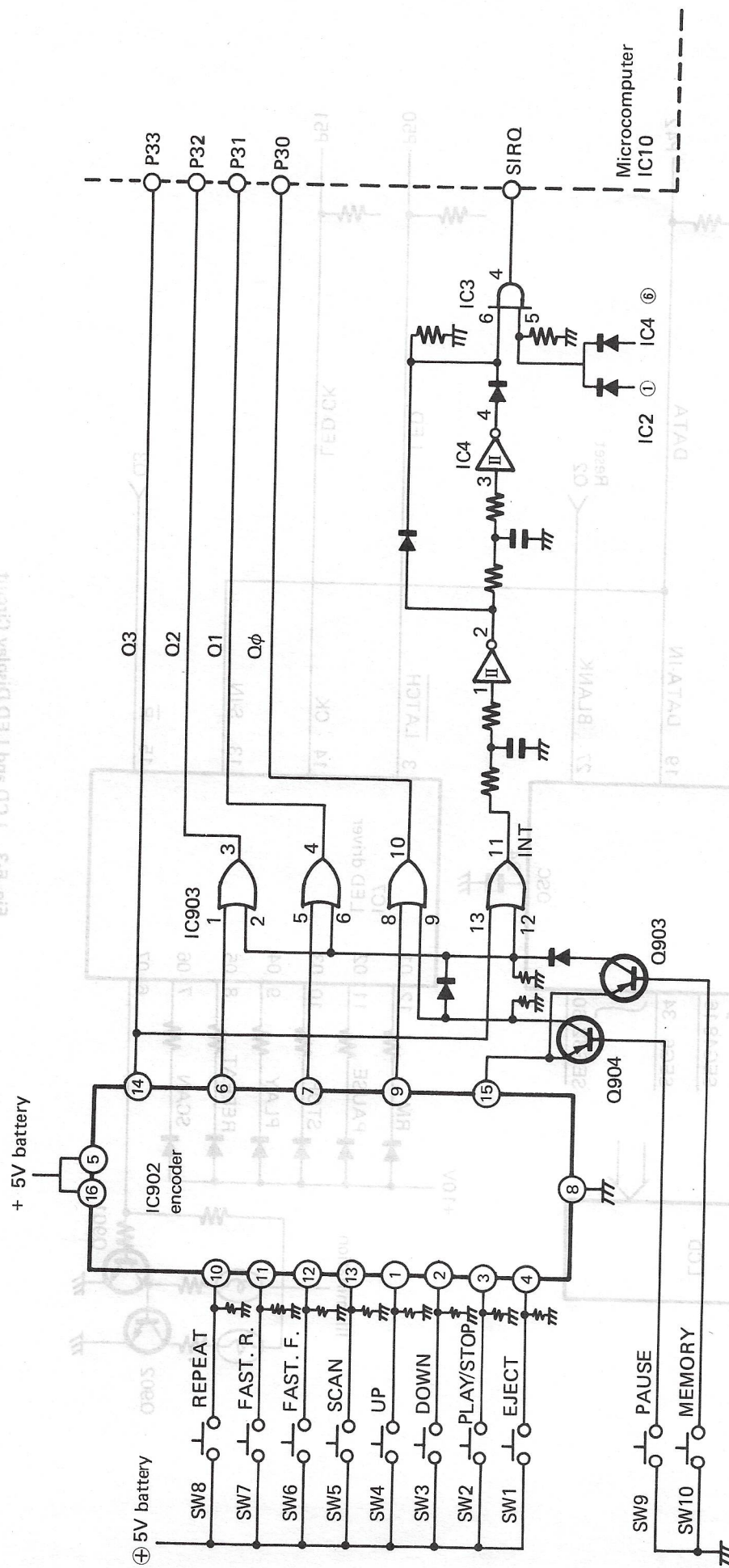


Fig. 6-1 AN8370S Block Diagram

5-2 Keyboard Interface Circuit



Ten key inputs are converted to 4-bit codes to give commands to the microcomputer. INT signal is routed through the delay circuit and waveform shaping circuit to supply SIRQ H pulse.

5-2 Keyboard Interface Circuit

5-3 LCD and LED Display Circuit

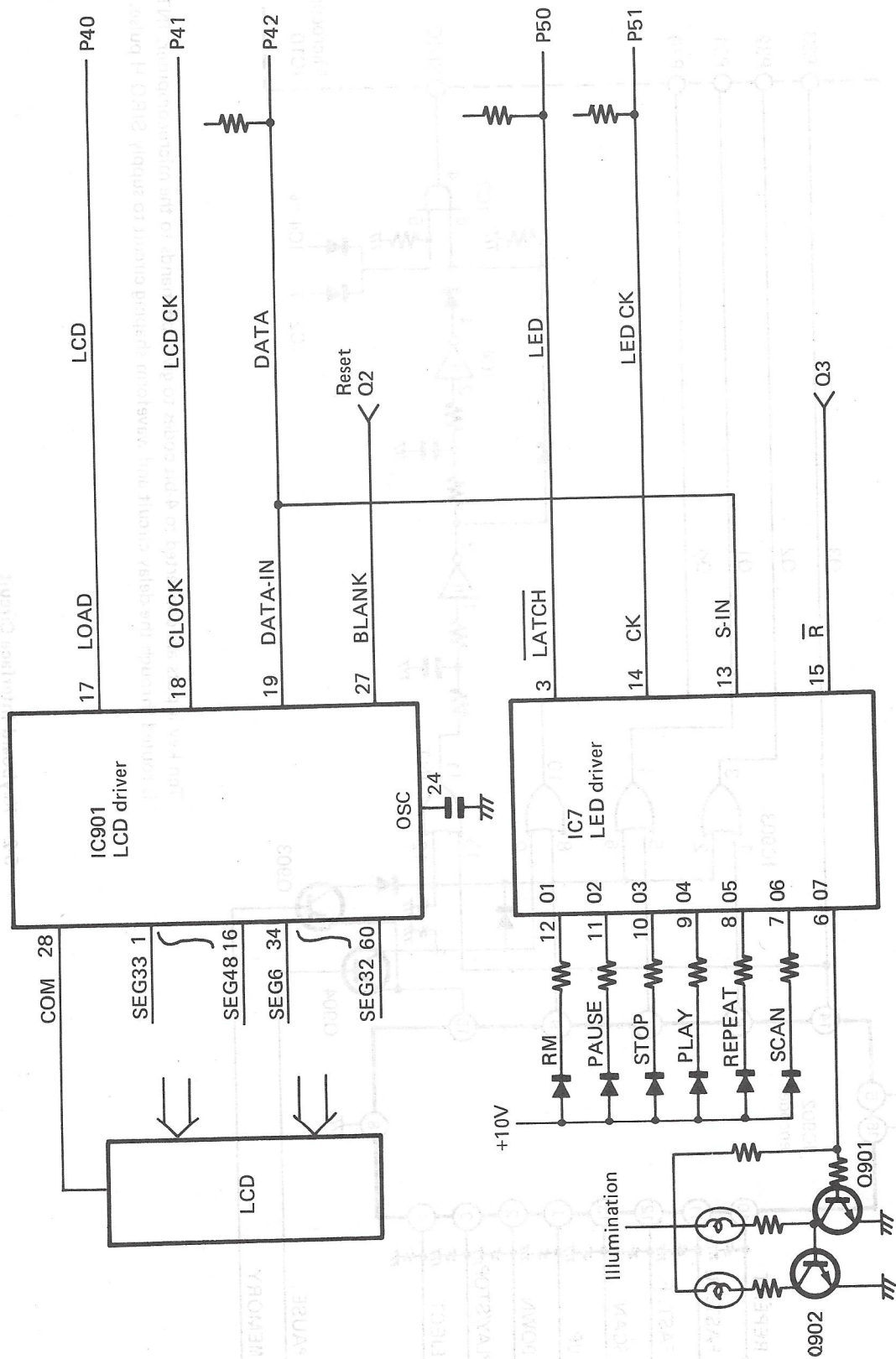


Fig. 5-3 LCD and LED Display Circuit

5. MICROCOMPUTER PERIPHERAL CIRCUITS

5-1 Reset Circuit

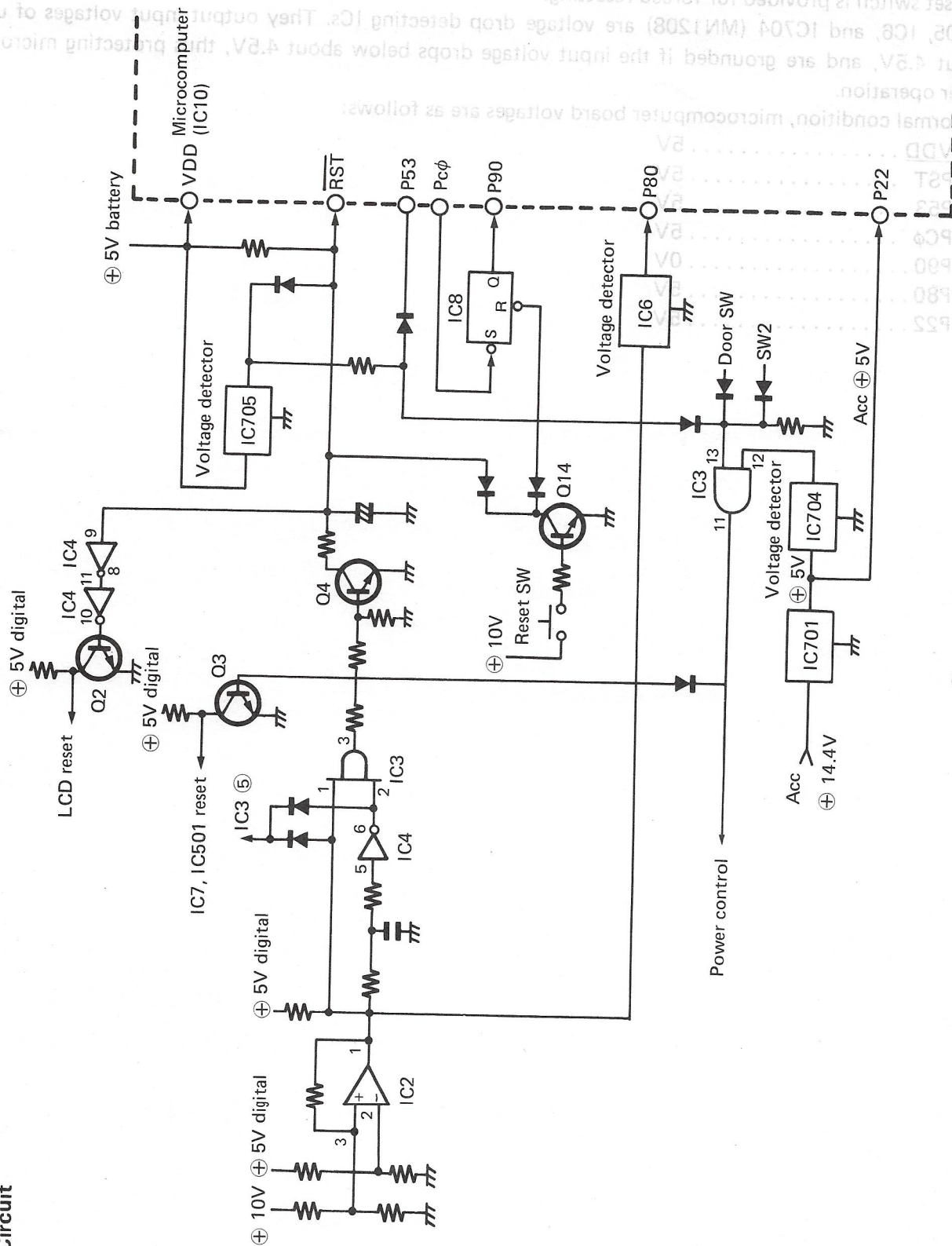


Fig. 5-1 Reset Circuit

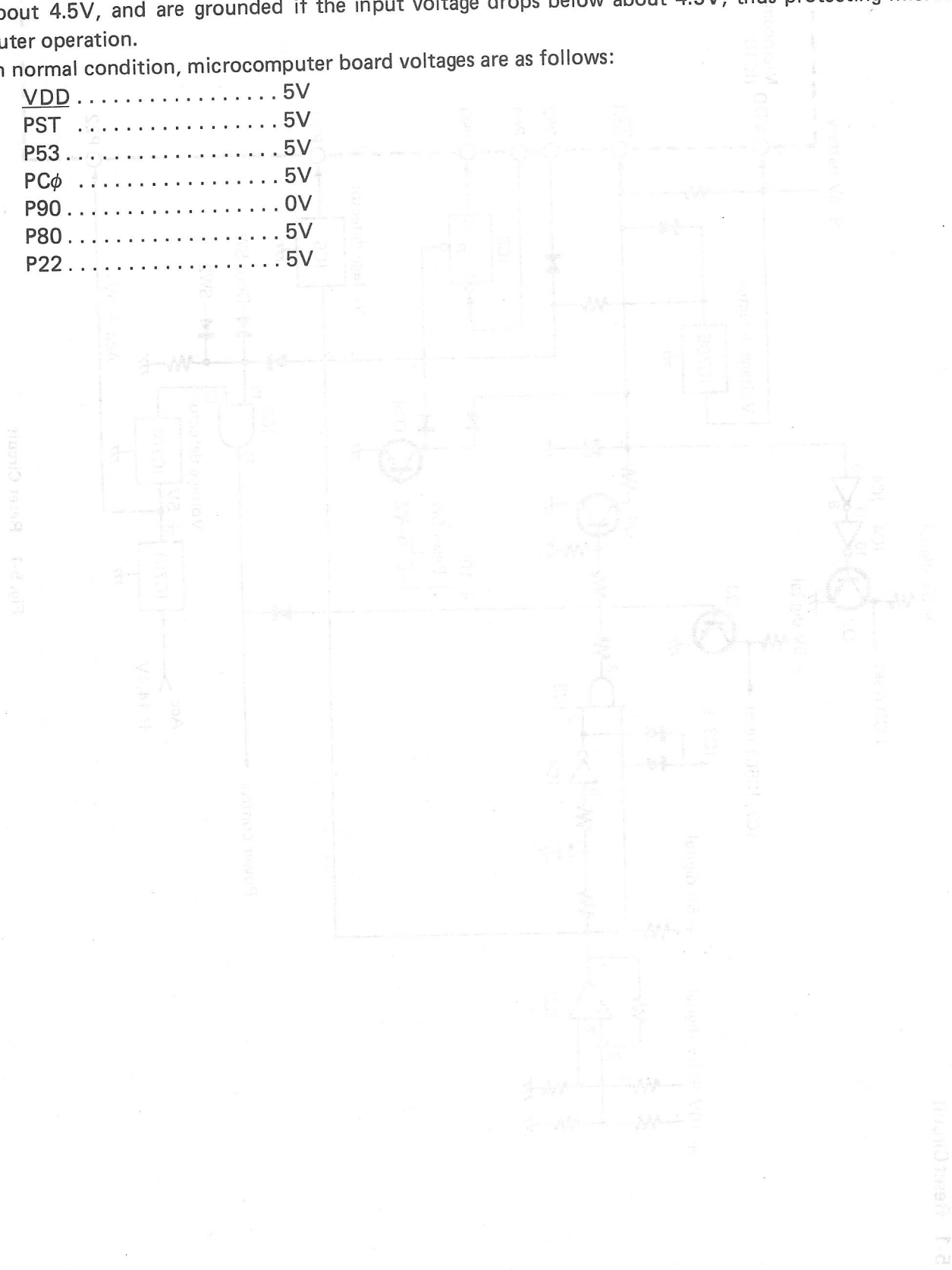
The reset circuit is so designed that it will generate a reset pulse when a disk is inserted. IC2 detects a rising voltage, and IC4 and IC3 generate a reset pulse.

A reset switch is provided for forced resetting.

IC705, IC6, and IC704 (MN1208) are voltage drop detecting ICs. They output input voltages of up to about 4.5V, and are grounded if the input voltage drops below about 4.5V, thus protecting microcomputer operation.

In normal condition, microcomputer board voltages are as follows:

VDD	5V
PST	5V
P53	5V
PCφ	5V
P90	0V
P80	5V
P22	5V



4. POWER CIRCUIT

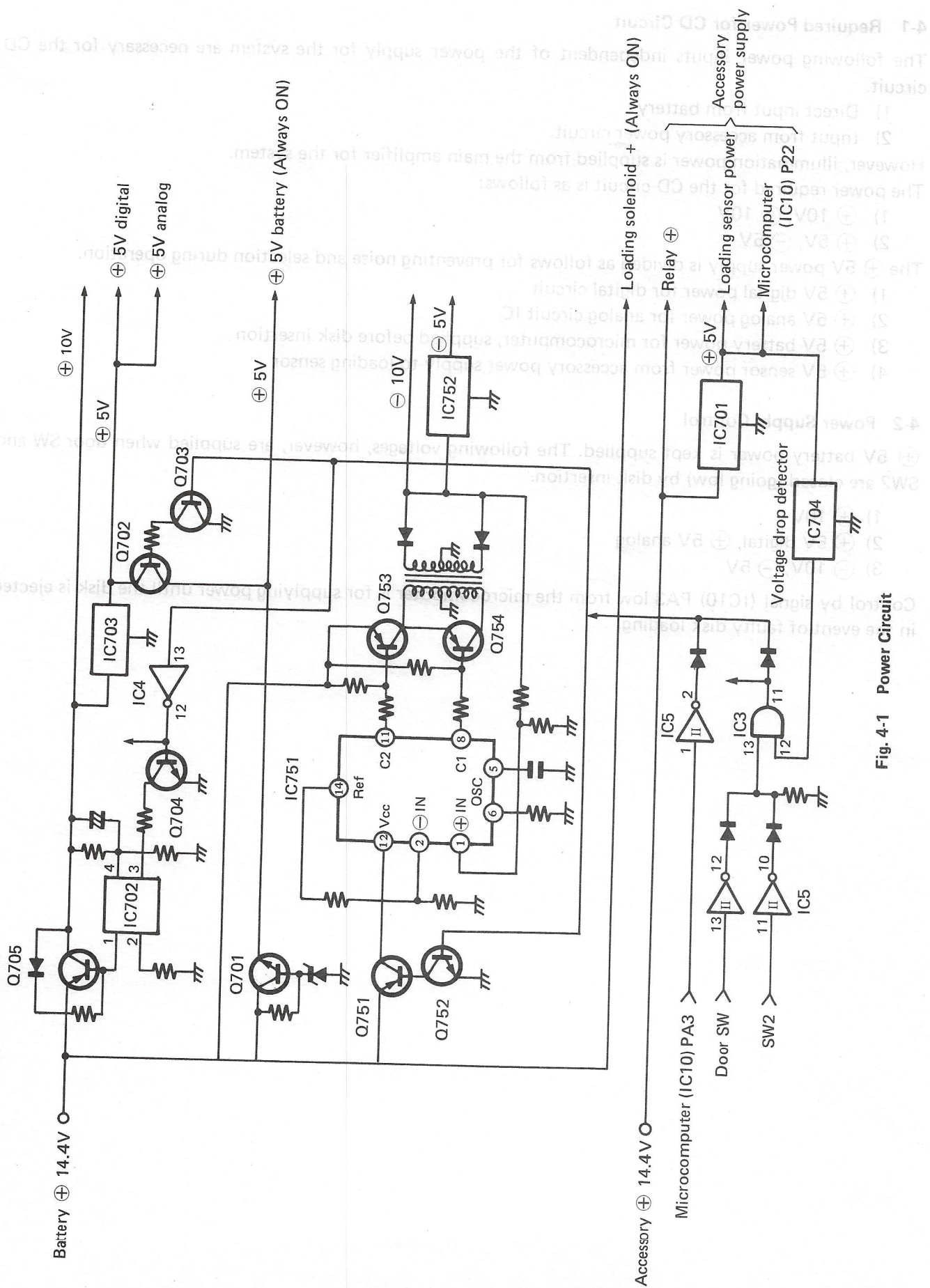


Fig. 4-1 Power Circuit

4-1 Required Power for CD Circuit

The following power inputs independent of the power supply for the system are necessary for the CD circuit.

- 1) Direct input from battery
- 2) Input from accessory power circuit

However, illumination power is supplied from the main amplifier for the system.

The power required for the CD circuit is as follows:

- 1) $\oplus 10V, \ominus 10V$
- 2) $\oplus 5V, \ominus 5V$

The $\oplus 5V$ power supply is divided as follows for preventing noise and selection during operation.

- 1) $\oplus 5V$ digital power for digital circuit
- 2) $\oplus 5V$ analog power for analog circuit IC
- 3) $\oplus 5V$ battery power for microcomputer, supplied before disk insertion
- 4) $\oplus 5V$ sensor power from accessory power supply to loading sensor

4-2 Power Supply Control

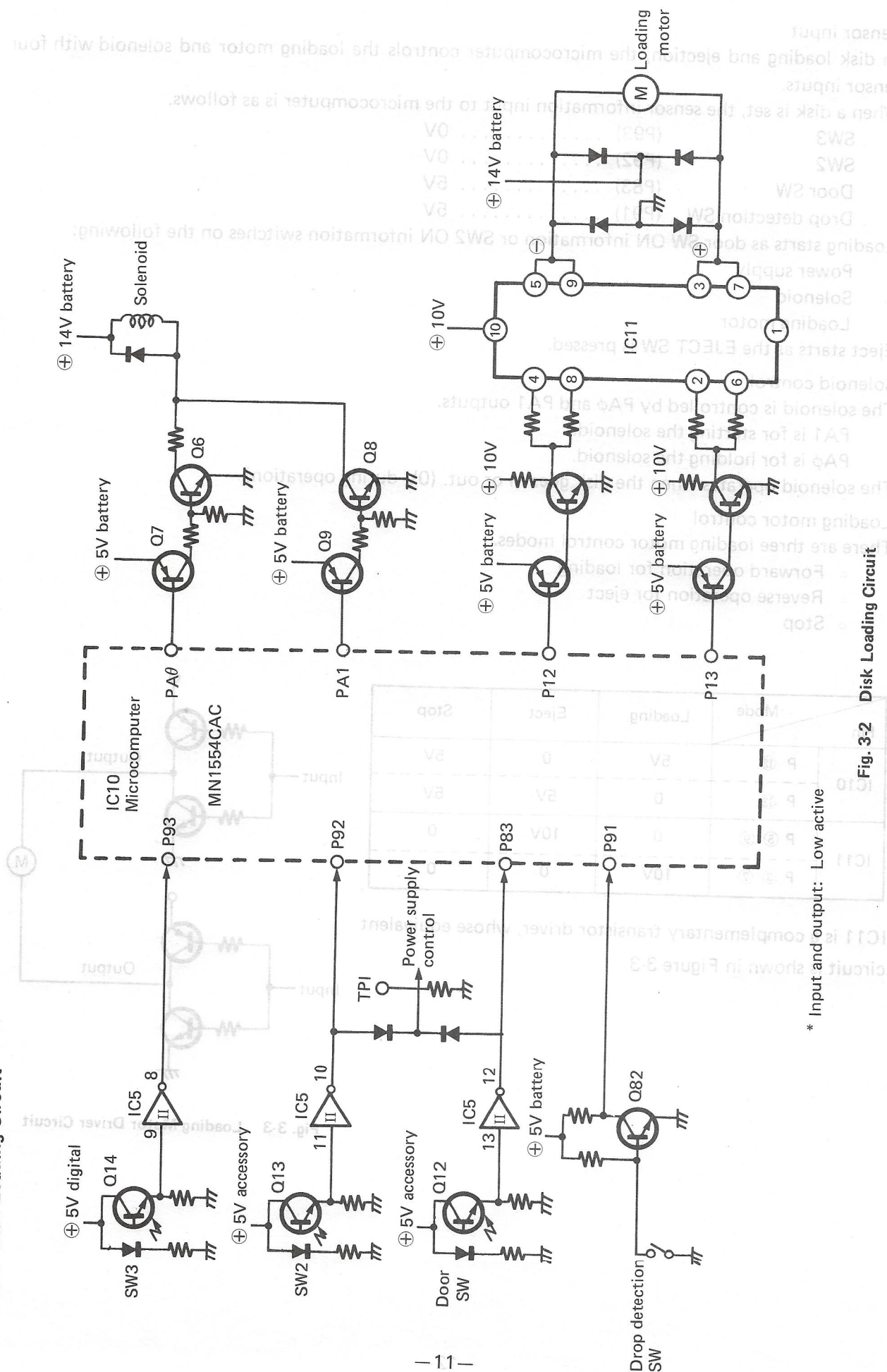
$\oplus 5V$ battery power is kept supplied. The following voltages, however, are supplied when door SW and SW2 are closed (going low) by disk insertion.

- 1) $\oplus 10V$
- 2) $\oplus 5V$ digital, $\oplus 5V$ analog
- 3) $\ominus 10V, \ominus 5V$

Control by signal (IC10) PA3 low from the microcomputer is for supplying power until the disk is ejected in the event of faulty disk loading.



3-4 Disk Loading Circuit



* Input and output: Low active

(1) Sensor input

In disk loading and ejection, the microcomputer controls the loading motor and solenoid with four sensor inputs.

When a disk is set, the sensor information input to the microcomputer is as follows.

SW3	(P93)		0V
SW2	(P92)		0V
Door SW	(P83)		5V
Drop detection SW	(P91)		5V

Loading starts as door SW ON information or SW2 ON information switches on the following:

- Power supply
- Solenoid
- Loading motor

Eject starts as the EJECT SW is pressed.

(2) Solenoid control

The solenoid is controlled by PA ϕ and PA1 outputs.

PA1 is for starting the solenoid.

PA ϕ is for holding the solenoid.

The solenoid operates when the disk goes in or out. (0V during operation)

(3) Loading motor control

There are three loading motor control modes.

- Forward operation for loading
- Reverse operation for eject
- Stop

Pin \ Mode		Loading	Eject	Stop
IC10	P ⑫	5V	0	5V
	P ⑬	0	5V	5V
IC11	P ⑤ ⑨	0	10V	0
	P ③ ⑦	10V	0	0

IC11 is a complementary transistor driver, whose equivalent circuit is shown in Figure 3-3.

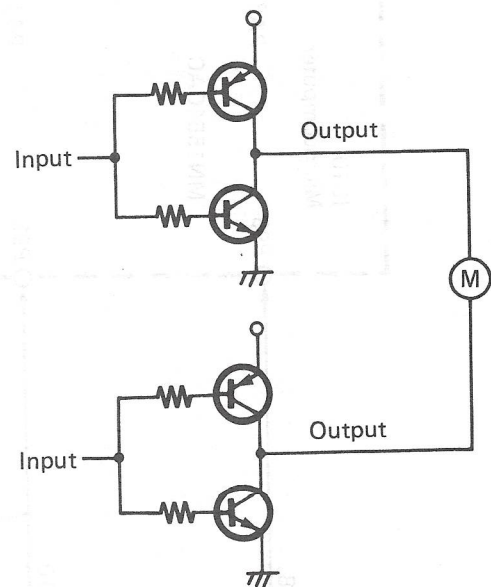
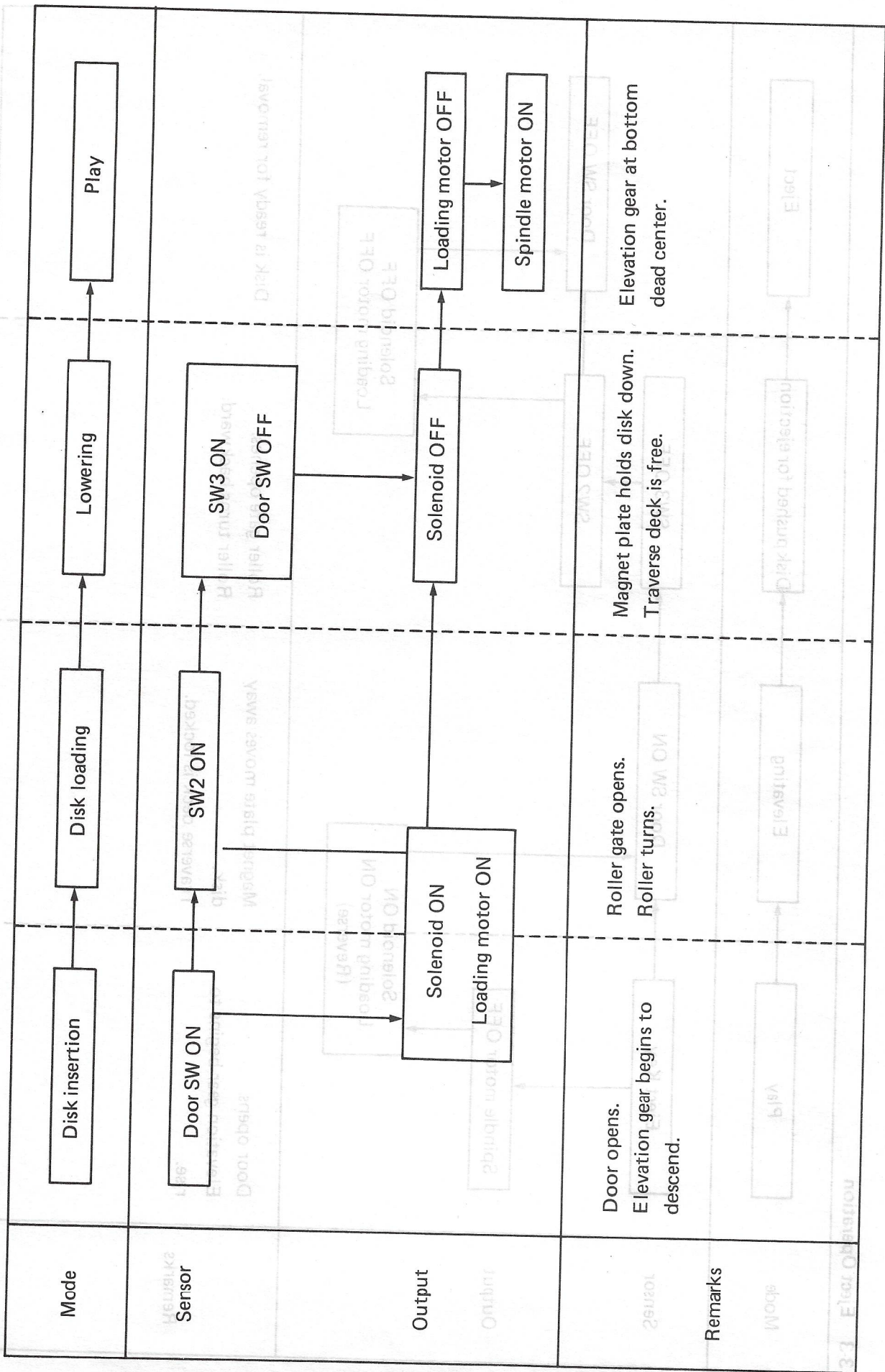


Fig. 3-3 Loading Motor Driver Circuit

3-2 Loading Operation



3-3 Eject Operation

